Capote 2026-05-04 Fit Chain: Step-by-Step Benchmark Report

Round 2 — incorporates corrections requested in the 2026-05-08 email round (see `references/emails/tnc_update_roberto_20260508/`).

This report documents the implementation of the new TNC fit chain proposed in the 2026-05-04 emails *"NEW TNC FIT steps"* and *"Revision/additions to Axton Total cross sections experimental data"*, refined by the corrections in the 2026-05-08 round (drop KAISER as duplicate of ARIF; reclassify ATOYI / WILLIS / FOOTE / GREEN-233 from SCA to SCR; replace Westcott factors with B81-derived values). Each section corresponds to one of the seven steps (0–6) in the chain, showing how the iterative Generalized Least Squares (GLS) fit results evolve as data are added, deleted, reclassified, or modified.

All fits use the microscopic-only data subset (Maxwellian-derived functions excluded) on the Axton 1986 database. Steps 0 and 1 read from a round-2 baseline file (`axton1986_data_capote_round2_baseline.json`) that is identical to the original microscopic-Axton baseline except for the new Westcott factors; the original `axton1986_data.json` is preserved unchanged as the Axton-Table-8 reproduction artefact.

Quantities with shifts exceeding 0.1% relative to the previous step are highlighted in **bold**. The Westcott-folded quantities FA, FF, F1ETA, F2ETA, F3ETA are computed as derived outputs at each step per the request (K1 = F3ETA). A Round-1 → Round-2 comparison table is shown after the Step-5-vs-Step-6 head-to-head; the round-2 corrections are documented in detail in Appendix B.

Overview

#	Step	N meas	N free	DOF	χ^2	Converged
0	Axton T8 baseline (no Lounsbury)	102	30	72	55.13	Yes
1	+ Lounsbury alpha (21–23)	105	30	75	60.49	Yes
2	+ Adamchuk 1988 alpha(^{235}U)	106	30	76	60.56	Yes
3	+ Wallner 2014 U-235 capture	108	30	78	62.85	Yes
4	Sears SCA list + Pu-241 estimate; delete Lemmel SCA	117	30	87	70.91	Yes
5	+ transmission additions; 3 SCR→SCA reclassifications	127	30	97	80.55	Yes
6	All ABS+SCR → ABS+SCA; drop Lemmel SCA-SCR; SCR partially removed	123	28	95	80.14	Yes

Targeted comparison: Step 5 vs Step 6

The 2026-05-04 email asks to *"compare steps 5 and 6 — hopefully one of them will be the final fit"*. The two candidate fits agree to better than 0.2% on every primary and derived quantity. Step 5 retains 4 SCR fit variables (now constrained by both the round-2 SCR reclassifications and the ABS+SCR transmissions); step 6 removes the ABS+SCR transmissions and Lemmel SCA-SCR, leaving SCR(^{233}U) and SCR(^{235}U) constrained only by the reclassified direct measurements (and SCR(^{239}Pu), SCR(^{241}Pu) unconstrained → removed). The 2026-05-06 follow-up email expresses a marginal preference for Step 5 over Step 6.

Step 0: Axton Table 8 Baseline (No Lounsbury)

This is the reference point: the microscopic GLS fit reproducing Axton's Table 8 results, with Lounsbury alpha measurements (Nos. 21–23) excluded. The detailed reproducibility analysis and the rationale for excluding Lounsbury are documented in `report_axton1986_reproduction.md` (Section 4); briefly, Lounsbury alpha measurements carry a "Z" systematic code (*Lounsbury Temperature Uncertainty*) that makes them effectively Maxwellian/integral, and Axton excluded them in his Table 6/Table 8 microscopic fit.

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	12.3206	5.681	–
SCA 35	16.2235	7.385	–
SCA 39	7.9001	12.380	–
SCA 41	11.9720	21.596	–
SCR 33	11.1483	8.161	–
SCR 35	14.5068	9.455	–
SCR 39	6.8449	24.890	–
SCR 41	10.8234	35.520	–
ABS 33	575.2413	0.321	–
ABS 35	681.2060	0.249	–
ABS 39	1018.7914	0.362	–
ABS 41	1382.2612	1.079	–
FIS 33	533.1364	0.460	–
FIS 35	585.0334	0.277	–
FIS 39	748.5245	0.340	–
FIS 41	1020.3307	1.127	–
CA 33	42.1049	4.228	–
CA 35	96.1726	1.811	–
CA 39	270.2669	1.181	–
CA 41	361.9304	1.729	–
CAP 34	95.9000	2.086	–
NUB 33	2.4860	0.217	–
NUB 35	2.4264	0.188	–
NUB 39	2.8794	0.207	–
NUB 41	2.9406	0.220	–
NUB 52	3.7649	0.132	–
ETA 33	2.3040	0.269	–
ETA 35	2.0838	0.255	–
ETA 39	2.1156	0.310	–
ETA 41	2.1706	0.399	–
ALPHA 33	0.0790	4.545	–
ALPHA 35	0.1644	1.962	–
ALPHA 39	0.3611	1.289	–
ALPHA 41	0.3547	1.697	–
WGA 33	1.0000	0.100	–
WGA 35	0.9820	0.305	–
WGA 39	1.0780	0.928	–
WGA 41	1.0460	0.669	–
WGF 33	0.9970	0.100	–
WGF 35	0.9810	0.510	–
WGF 39	1.0520	1.331	–
WGF 41	1.0520	0.665	–
FA 33	575.2413	0.337	–
FA 35	668.9443	0.394	–
FA 39	1098.2571	0.996	–
FA 41	1445.8452	1.270	–
FF 33	531.5370	0.471	–
FF 35	573.9178	0.580	–
FF 39	787.4478	1.374	–
FF 41	1073.3879	1.309	–
F1ETA 33	2.2971	0.304	–
F1ETA 35	2.0817	0.647	–
F1ETA 39	2.0645	1.652	–
F1ETA 41	2.1831	1.025	–
F2ETA 33	1321.3993	0.429	–
F2ETA 35	1392.5407	0.586	–
F2ETA 39	2267.4064	1.379	–
F2ETA 41	3156.3904	1.301	–
F3ETA 33	746.1581	0.604	–
F3ETA 35	723.5963	1.123	–
F3ETA 39	1169.1493	2.779	–
F3ETA 41	1710.5452	1.844	–

Step 1: + Lounsbury Alpha (Nos. 21–23)

Step 1 adds the three Lounsbury 1970 alpha measurements back into the microscopic fit:

- No.21 LOUNSBURY (CA 33)/FIS 33 = 0.0861, σ = 2.24%
- No.22 LOUNSBURY (CA 35)/FIS 35 = 0.1697, σ = 1.71%
- No.23 LOUNSBURY (CA 39)/FIS 39 = 0.3555, σ = 1.60%

This is the largest single change in the entire chain. Lounsbury's $\alpha(^{233}U) = 0.0861$ is far above Axton's converged value of 0.0787, and pulling the system toward this value drives a large redistribution of ABS(^{233}U) between capture and fission — ALPHA(^{233}U) shifts by +7.2%, CA(^{233}U) by +6.6%, and FIS(^{233}U) drops by 0.5%. F3ETA(^{233}U) drops by 0.56%.

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	12.3166	5.681	-0.032
SCA 35	**15.8847**	7.270	**-2.088**
SCA 39	**7.9230**	12.364	**+0.291**
SCA 41	11.9618	21.596	-0.085
SCR 33	11.1419	8.171	-0.057
SCR 35	**14.0797**	9.527	**-2.944**
SCR 39	**6.9469**	23.295	**+1.490**
SCR 41	**10.8049**	35.707	**-0.170**
ABS 33	575.2692	0.321	+0.005
ABS 35	**682.0852**	0.233	**+0.129**
ABS 39	1018.2345	0.340	-0.055
ABS 41	1382.7102	1.079	+0.032
FIS 33	**530.3843**	0.355	**-0.516**
FIS 35	**584.0124**	0.244	**-0.175**
FIS 39	748.1165	0.321	-0.055
FIS 41	**1018.4694**	1.119	**-0.182**
CA 33	**44.8849**	1.837	**+6.603**
CA 35	**98.0728**	1.152	**+1.976**
CA 39	270.1180	0.913	-0.055
CA 41	**364.2408**	1.663	**+0.638**
CAP 34	95.9000	2.086	+0.000
NUB 33	**2.4910**	0.184	**+0.203**
NUB 35	**2.4294**	0.172	**+0.123**
NUB 39	2.8814	0.192	+0.067
NUB 41	2.9435	0.209	+0.099
NUB 52	3.7664	0.129	+0.039
ETA 33	**2.2967**	0.199	**-0.319**
ETA 35	**2.0801**	0.211	**-0.180**
ETA 39	2.1170	0.267	+0.067
ETA 41	**2.1681**	0.391	**-0.116**
ALPHA 33	**0.0846**	1.960	**+7.156**
ALPHA 35	**0.1679**	1.235	**+2.154**
ALPHA 39	0.3611	0.965	-0.001
ALPHA 41	**0.3576**	1.585	**+0.822**
WGA 33	1.0000	0.100	+0.000
WGA 35	0.9820	0.305	+0.000
WGA 39	1.0780	0.928	+0.000
WGA 41	1.0460	0.669	+0.000
WGF 33	0.9970	0.100	+0.000
WGF 35	0.9810	0.510	+0.000
WGF 39	1.0520	1.331	+0.000
WGF 41	1.0520	0.665	+0.000
FA 33	575.2692	0.337	+0.005
FA 35	**669.8077**	0.384	**+0.129**
FA 39	1097.6568	0.988	-0.055
FA 41	1446.3148	1.270	+0.032
FF 33	**528.7932**	0.369	**-0.516**
FF 35	**572.9162**	0.565	**-0.175**
FF 39	787.0186	1.369	-0.055
FF 41	**1071.4298**	1.302	**-0.182**
F1ETA 33	**2.2898**	0.244	**-0.319**
F1ETA 35	**2.0779**	0.630	**-0.180**
F1ETA 39	2.0659	1.644	+0.067
F1ETA 41	**2.1805**	1.021	**-0.116**
F2ETA 33	**1317.2452**	0.391	**-0.314**
F2ETA 35	1391.8258	0.582	-0.051
F2ETA 39	2267.6891	1.378	+0.012
F2ETA 41	3153.7617	1.300	-0.083
F3ETA 33	**741.9759**	0.515	**-0.560**
F3ETA 35	**722.0181**	1.108	**-0.218**
F3ETA 39	1170.0323	2.771	+0.076
F3ETA 41	**1707.4469**	1.840	**-0.181**

Step 2: + Adamchuk 1988 Alpha(^{235}U)

Step 2 adds one new measurement (No.168):

- ADAMCHUK (CA 35)/FIS 35 = 0.169, σ = 2.1%, At. En. (R) 65, 434 (1988) (GMA NS=8026)

Adamchuk's α(^{235}U) = 0.169 is highly consistent with the step-1 fitted value (0.168), so the step is essentially a no-op for χ^2: only +0.07 from one added measurement, and shifts in U-235 quantities are at the 0.1–0.2% level. All other nuclides are unchanged.

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	12.3166	5.681	-0.000
SCA 35	**15.8609**	7.245	**-0.150**
SCA 39	7.9233	12.364	+0.003
SCA 41	11.9618	21.596	-0.000
SCR 33	11.1419	8.171	-0.000
SCR 35	**14.0487**	9.509	**-0.220**
SCR 39	6.9479	23.281	+0.014
SCR 41	10.8049	35.708	-0.000
ABS 33	575.2694	0.321	+0.000
ABS 35	682.1500	0.230	+0.010
ABS 39	1018.2347	0.340	+0.000
ABS 41	1382.7113	1.079	+0.000
FIS 33	530.3690	0.355	-0.003
FIS 35	583.9317	0.239	-0.014
FIS 39	748.0666	0.320	-0.007
FIS 41	1018.3865	1.119	-0.008
CA 33	44.9004	1.832	+0.035
CA 35	**98.2183**	0.999	**+0.148**
CA 39	270.1681	0.910	+0.019
CA 41	364.3248	1.660	+0.023
CAP 34	95.9000	2.086	+0.000
NUB 33	2.4911	0.184	+0.002
NUB 35	2.4295	0.170	+0.006
NUB 39	2.8815	0.191	+0.003
NUB 41	2.9436	0.208	+0.004
NUB 52	3.7664	0.129	+0.002
ETA 33	2.2967	0.199	-0.001
ETA 35	2.0797	0.200	-0.017
ETA 39	2.1169	0.266	-0.003
ETA 41	2.1680	0.390	-0.005
ALPHA 33	0.0847	1.955	+0.037
ALPHA 35	**0.1682**	1.064	**+0.162**
ALPHA 39	0.3612	0.960	+0.025
ALPHA 41	0.3577	1.580	+0.031
WGA 33	1.0000	0.100	+0.000
WGA 35	0.9820	0.305	+0.000
WGA 39	1.0780	0.928	+0.000
WGA 41	1.0460	0.669	+0.000
WGF 33	0.9970	0.100	+0.000
WGF 35	0.9810	0.510	+0.000
WGF 39	1.0520	1.331	+0.000
WGF 41	1.0520	0.665	+0.000
FA 33	575.2694	0.337	+0.000
FA 35	669.8713	0.382	+0.010
FA 39	1097.6570	0.988	+0.000
FA 41	1446.3160	1.270	+0.000
FF 33	528.7779	0.369	-0.003
FF 35	572.8370	0.563	-0.014
FF 39	786.9660	1.369	-0.007
FF 41	1071.3426	1.302	-0.008
F1ETA 33	2.2898	0.244	-0.001
F1ETA 35	2.0776	0.627	-0.017
F1ETA 39	2.0659	1.644	-0.003
F1ETA 41	2.1804	1.021	-0.005
F2ETA 33	1317.2344	0.391	-0.001
F2ETA 35	1391.7199	0.582	-0.008
F2ETA 39	2267.6139	1.378	-0.003
F2ETA 41	3153.6197	1.300	-0.005
F3ETA 33	741.9650	0.515	-0.001
F3ETA 35	721.8486	1.104	-0.023

Quantity	Value	Unc (%)	Δ vs prev (%)
F3ETA 39	1169.9569	2.771	-0.006
F3ETA 41	1707.3037	1.840	-0.008

Step 3: + Wallner 2014 U-235 Capture

Step 3 adds two Wallner et al. 2014 thermal capture measurements (PRL 112, 192501, 2014):

- No.169 WALLNER CA 35 = 99.5 b, σ = 2.0%
- No.170 WALLNER CA 35 = 101.5 b, σ = 2.0%

Both Wallner values lie above the step-2 fit (CA 35 ≈ 98.2 b). The fit accommodates the pull with χ^2 = +2.3 over 2 added measurements; CA(^{235}U) shifts +0.76% to 98.97 b. Through the absorption channel correlations, SCA(^{235}U) drops 1.3% and SCR(^{235}U) drops 1.9%. F3ETA(^{235}U) shifts only 0.08%, so the K1 of interest is essentially preserved.

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	12.3165	5.681	-0.001
SCA 35	**15.6565**	7.185	**-1.288**
SCA 39	7.9214	12.364	-0.024
SCA 41	11.9602	21.596	-0.013
SCR 33	11.1417	8.172	-0.002
SCR 35	**13.7764**	9.597	**-1.938**
SCR 39	**6.9407**	23.380	**-0.103**
SCR 41	10.8020	35.738	-0.027
ABS 33	575.2702	0.321	+0.000
ABS 35	682.7209	0.220	+0.084
ABS 39	1018.3508	0.340	+0.011
ABS 41	1382.7815	1.079	+0.005
FIS 33	530.2947	0.355	-0.014
FIS 35	583.7524	0.237	-0.031
FIS 39	747.9269	0.320	-0.019
FIS 41	1018.0396	1.119	-0.034
CA 33	**44.9755**	1.826	**+0.167**
CA 35	**98.9685**	0.814	**+0.764**
CA 39	270.4239	0.907	+0.095
CA 41	**364.7420**	1.657	**+0.114**
CAP 34	95.9000	2.086	+0.000
NUB 33	2.4913	0.184	+0.010
NUB 35	2.4303	0.169	+0.030
NUB 39	2.8819	0.191	+0.016
NUB 41	2.9441	0.208	+0.018
NUB 52	3.7667	0.129	+0.007
ETA 33	2.2966	0.199	-0.004
ETA 35	2.0780	0.190	-0.084
ETA 39	2.1166	0.266	-0.014
ETA 41	2.1675	0.390	-0.022
ALPHA 33	**0.0848**	1.949	**+0.181**
ALPHA 35	**0.1695**	0.879	**+0.795**
ALPHA 39	**0.3616**	0.956	**+0.113**
ALPHA 41	**0.3583**	1.574	**+0.149**
WGA 33	1.0000	0.100	+0.000
WGA 35	0.9820	0.305	+0.000
WGA 39	1.0780	0.928	+0.000
WGA 41	1.0460	0.669	+0.000
WGF 33	0.9970	0.100	+0.000
WGF 35	0.9810	0.510	+0.000
WGF 39	1.0520	1.331	+0.000
WGF 41	1.0520	0.665	+0.000
FA 33	575.2702	0.337	+0.000
FA 35	670.4319	0.377	+0.084
FA 39	1097.7822	0.988	+0.011
FA 41	1446.3895	1.270	+0.005
FF 33	528.7038	0.369	-0.014
FF 35	572.6611	0.562	-0.031
FF 39	786.8191	1.369	-0.019
FF 41	1070.9776	1.302	-0.034
F1ETA 33	2.2897	0.244	-0.004

Quantity	Value	Unc (%)	Δ vs prev (%)
F1ETA 35	2.0758	0.624	-0.084
F1ETA 39	2.0656	1.644	-0.014
F1ETA 41	2.1800	1.021	-0.022
F2ETA 33	1317.1830	0.391	-0.004
F2ETA 35	1391.7155	0.581	-0.000
F2ETA 39	2267.5523	1.378	-0.003
F2ETA 41	3153.0973	1.300	-0.017
F3ETA 33	741.9129	0.515	-0.007
F3ETA 35	721.2836	1.104	-0.078
F3ETA 39	1169.7702	2.771	-0.016
F3ETA 41	1706.7079	1.840	-0.035

Step 4: Sears SCA List + Pu-241 Estimate; Delete Lemmel SCA

Step 4 implements the most extensive change in the chain. It replaces the four Lemmel direct SCA measurements with a curated compilation drawn from Sears (1992), the Divadeenam compilation, EXFOR replacements, and Pronyaev's potential-scattering estimate for Pu-241.

The Lemmel SCA–SCR datapoints (Nos. 89, 91, 93, 95) are *kept* per the instruction *"leave SCA-SCR"*; only the four direct Lemmel SCA datapoints are deleted.

Round-2 corrections applied here (see Appendix B): KAISER (No.173) is removed as a duplicate of ARIF; ATOYI (No.171), WILLIS (No.172), FOOTE (No.177) and GREEN-233 (No.179) are reclassified from SCA to SCR. Net step-4 cardinality is therefore 9 added (vs. 14 before round 2): 1 deleted (Kaiser) plus 4 reclassified (still added, but now constraining SCR rather than SCA).

Datapoints deleted

No.	Author	Reference	Quantity	Value	Note
88	Lemmel	1982	SCA(^{233}U)	12.306	replaced by step-4 SCA list
90	Lemmel	1982	SCA(^{235}U)	16.508	replaced by step-4 SCA list
92	Lemmel	1982	SCA(^{239}Pu)	8.004	replaced by step-4 SCA list
94	Lemmel	1982	SCA(^{241}Pu)	12.006	replaced by Pronyaev 12.0(0.8)

Datapoints added

No.	Author / year	Reference	Type	Value (b)	σ (b)	Appendix
171	Atoyi	Acta Cryst. 20, 587 (1966)	SCR(^{235}U) [r2]	12.3	0.5	B2
172	Willis	PRS A274, 122 (1963)	SCR(^{235}U) [r2]	12.8	1	B2
173	Kaiser	Physica B 136, 134 (1986)	— DELETED [r2]	14	0.2	B1
174	Arif	PR A35, 2810 (1987)	SCA(^{235}U)	14	0.2	
175	Ceulemans	Divadeenam compilation 1970	SCA(^{235}U)	14.6	0.5	A4
176	Bayvas	Divadeenam compilation 1980	SCA(^{235}U)	14.3	1.5	
177	Foote	Divadeenam compilation 1958	SCR(^{235}U) [r2]	15	1	B3
178	Oleksa	Phys Rev 109, 1645 (1958)	SCA(^{233}U)	13.4	0.5	A5
179	Green	JNM 51, 281 (1974)	SCR(^{233}U) [r2]	12.6	0.7	B2
180	Vertebnyi	Divadeenam compilation 1974	SCA(^{233}U)	13.2	0.6	
181	Green	JNM 34, 281 (1970)	SCA(^{239}Pu)	7.7	0.6	
182	Atoyi	Acta Cryst. 20, 587 (1966)	SCA(^{239}Pu)	7.3	0.7	
183	Roof	Acta Cryst. 15, 351 (1962)	SCA(^{239}Pu)	7.3	0.7	
184	Pronyaev	Internal analysis 2026-04-17	SCA(^{241}Pu)	12	0.8	A7

Impact The U-235 SCA channel is the most affected: SCA(^{235}U) drops driven by the ARIF interferometry value 14.0 b replacing the 16-ish Lemmel value (Kaiser is no longer added separately; it is the same experiment as Arif). SCA uncertainties collapse dramatically, reflecting the data richness of the new compilation. ABS, FIS, CA, and the F-quantities shift by ≤0.2% relative to step 3. χ^2 rises by ≈8 from a net +9 measurements (~0.9/measurement, indicating moderate but absorbable tension).

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	**13.3989**	2.631	**+8.789**
SCA 35	**14.1217**	1.265	**-9.803**

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 39	**7.4596**	5.107	**-5.829**
SCA 41	**11.9951**	6.665	**+0.292**
SCR 33	**12.3565**	3.809	**+10.902**
SCR 35	**12.7854**	2.629	**-7.193**
SCR 39	**6.4694**	21.374	**-6.791**
SCR 41	**10.8324**	27.684	**+0.282**
ABS 33	**574.0827**	0.295	**-0.206**
ABS 35	**683.5588**	0.184	**+0.123**
ABS 39	1018.8655	0.336	+0.051
ABS 41	1382.9760	1.070	+0.014
FIS 33	**529.1971**	0.331	**-0.207**
FIS 35	**584.4090**	0.210	**+0.112**
FIS 39	748.3107	0.317	+0.051
FIS 41	1018.1747	1.111	+0.013
CA 33	**44.8856**	1.822	**-0.200**
CA 35	**99.1499**	0.793	**+0.183**
CA 39	270.5548	0.903	+0.048
CA 41	364.8014	1.650	+0.016
CAP 34	95.9000	2.086	+0.000
NUB 33	2.4914	0.184	+0.001
NUB 35	2.4303	0.169	+0.003
NUB 39	2.8819	0.191	+0.001
NUB 41	2.9442	0.208	+0.001
NUB 52	3.7667	0.129	+0.001
ETA 33	2.2966	0.199	+0.000
ETA 35	2.0778	0.190	-0.008
ETA 39	2.1167	0.266	+0.002
ETA 41	2.1676	0.390	+0.000
ALPHA 33	0.0848	1.949	+0.007
ALPHA 35	0.1697	0.875	+0.071
ALPHA 39	0.3616	0.954	-0.003
ALPHA 41	0.3583	1.574	+0.003
WGA 33	1.0000	0.100	+0.000
WGA 35	0.9820	0.305	+0.000
WGA 39	1.0780	0.928	+0.000
WGA 41	1.0460	0.669	+0.000
WGF 33	0.9970	0.100	+0.000
WGF 35	0.9810	0.510	+0.000
WGF 39	1.0520	1.331	+0.000
WGF 41	1.0520	0.665	+0.000
FA 33	**574.0827**	0.311	**-0.206**
FA 35	**671.2548**	0.357	**+0.123**
FA 39	1098.3370	0.987	+0.051
FA 41	1446.5929	1.262	+0.014
FF 33	**527.6095**	0.346	**-0.207**
FF 35	**573.3052**	0.551	**+0.112**
FF 39	787.2228	1.368	+0.051
FF 41	1071.1197	1.295	+0.013
F1ETA 33	2.2897	0.244	+0.000
F1ETA 35	2.0757	0.624	-0.008
F1ETA 39	2.0656	1.644	+0.002
F1ETA 41	2.1800	1.021	+0.000
F2ETA 33	**1314.4669**	0.369	**-0.206**
F2ETA 35	**1393.3184**	0.570	**+0.115**
F2ETA 39	2268.7368	1.377	+0.052
F2ETA 41	3153.5496	1.293	+0.014
F3ETA 33	**740.3842**	0.499	**-0.206**
F3ETA 35	**722.0637**	1.099	**+0.108**
F3ETA 39	1170.3998	2.771	+0.054
F3ETA 41	1706.9566	1.836	+0.015

Step 5: + Transmission Entries; 3 ABS+SCR → ABS+SCA Reclassifications

Step 5 implements Roberto's transmission-data revision from his "Revision/additions to Axton Total cross sections experimental data" email of 2026-05-04. Three existing transmission entries are reclassified, ten new entries are added, and two would-be additions (Spencer 1987) are recognised as already in the database.

Datapoints reclassified (ABS+SCR → ABS+SCA)

No.	Author	Year	Nuclide	Change	Appendix
68	Safford	1959	U-235	ABS+SCR → ABS+SCA	
78	Nikitin	1955	U-233	ABS+SCR → ABS+SCA	A8
83	Safford	1960	U-233	ABS+SCR → ABS+SCA	A9

Datapoints added

No.	Author / year	Reference	Type	Value (b)	σ (b)	Appendix
185	Antonov	1986	ABS+SCA(^{235}U)	695	15	A10
186	Gerasimov	1962	ABS+SCR(^{235}U)	685	9	A11
187	Leonard	1954	ABS+SCR(^{235}U)	702	14	A12
188	Pschenichnyi	1978	ABS+SCR(^{233}U)	587.9	3	A13
189	Harvey-1	1979	ABS+SCR(^{233}U)	585	1.4	A14
190	Harvey-2	1979	ABS+SCR(^{233}U)	592	6	A14
191	Schwartz	1958	ABS+SCR(^{241}Pu)	1410	80	
192	Anderson	1945	ABS+SCR(^{239}Pu)	1045	25	
193	Palevsky	1955	ABS+SCR(^{239}Pu)	1025	10	
194	Leonard	1956	ABS+SCR(^{239}Pu)	1055	13	A15

Datapoints not added (already in DB)

Author / year	Type	Value	σ (b)	Reason
Spencer 1987	(^{235}U) ABS+SCR	690	5	Already in DB as No.166 SPENCER 1985(3) val=690.0 σ=1.0% — values match within rounding (per project decision); uncertainty discrepancy logged in A1
Spencer 1987	(^{239}Pu) ABS+SCR	1025	6	Already in DB as No.167 SPENCER 1985(3) val=1025.8 σ=1.0% — values match within rounding; uncertainty discrepancy logged in A1

Impact The new transmission datapoints sharpen ABS (and through it FIS and CA) but barely touch the SCA channel from step 4. ABS uncertainties tighten by 10–35% across the four nuclides. The largest individual tension is **Leonard 56 ABS+SCR(^{239}Pu) = 1055 b** at +2.1σ from the predicted 1027 b; this contributes ≈4.5 to the χ^2 increase of 9.5. K1 = F3ETA shifts ≤0.21% across all four nuclides relative to step 4.

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	13.3908	2.624	-0.061
SCA 35	14.1213	1.265	-0.003
SCA 39	**7.4728**	5.106	**+0.176**
SCA 41	11.9951	6.665	-0.000
SCR 33	**12.3795**	3.744	**+0.186**
SCR 35	12.7734	2.636	-0.094
SCR 39	**6.6595**	17.420	**+2.939**
SCR 41	10.8374	27.557	+0.046
ABS 33	**573.4874**	0.186	**-0.104**
ABS 35	683.4317	0.181	-0.019
ABS 39	**1020.6859**	0.307	**+0.179**
ABS 41	1383.6990	1.053	+0.052
FIS 33	**528.6252**	0.240	**-0.108**
FIS 35	584.3285	0.207	-0.014
FIS 39	**749.1488**	0.304	**+0.112**
FIS 41	1018.5549	1.096	+0.037
CA 33	44.8622	1.806	-0.052
CA 35	99.1032	0.791	-0.047
CA 39	**271.5372**	0.863	**+0.363**
CA 41	365.1441	1.635	+0.094

Quantity	Value	Unc (%)	Δ vs prev (%)
CAP 34	95.9000	2.086	+0.000
NUB 33	2.4914	0.184	+0.003
NUB 35	2.4304	0.169	+0.002
NUB 39	2.8824	0.191	+0.016
NUB 41	2.9444	0.208	+0.008
NUB 52	3.7668	0.129	+0.002
ETA 33	2.2965	0.199	-0.001
ETA 35	2.0780	0.190	+0.007
ETA 39	2.1156	0.264	-0.051
ETA 41	2.1674	0.390	-0.007
ALPHA 33	0.0849	1.948	+0.056
ALPHA 35	0.1696	0.875	-0.033
ALPHA 39	**0.3625**	0.937	**+0.251**
ALPHA 41	0.3585	1.573	+0.057
WGA 33	1.0000	0.100	+0.000
WGA 35	0.9820	0.305	+0.000
WGA 39	1.0780	0.928	+0.000
WGA 41	1.0460	0.669	+0.000
WGF 33	0.9970	0.100	+0.000
WGF 35	0.9810	0.510	+0.000
WGF 39	1.0520	1.331	+0.000
WGF 41	1.0520	0.665	+0.000
FA 33	**573.4874**	0.212	**-0.104**
FA 35	671.1299	0.355	-0.019
FA 39	**1100.2995**	0.977	**+0.179**
FA 41	1447.3492	1.248	+0.052
FF 33	**527.0393**	0.261	**-0.108**
FF 35	573.2262	0.550	-0.014
FF 39	**788.1045**	1.365	**+0.112**
FF 41	1071.5198	1.282	+0.037
F1ETA 33	2.2897	0.244	-0.001
F1ETA 35	2.0758	0.624	+0.007
F1ETA 39	2.0646	1.643	-0.051
F1ETA 41	2.1798	1.021	-0.007
F2ETA 33	**1313.0907**	0.290	**-0.105**
F2ETA 35	1393.1589	0.569	-0.011
F2ETA 39	**2271.6351**	1.373	**+0.128**
F2ETA 41	3154.9640	1.279	+0.045
F3ETA 33	**739.6033**	0.444	**-0.105**
F3ETA 35	722.0290	1.098	-0.005
F3ETA 39	1171.3357	2.772	+0.080
F3ETA 41	1707.6149	1.827	+0.039

Step 6: All ABS+SCR → ABS+SCA; Drop Lemmel SCA-SCR; SCR Partially Removed

Step 6 finalises the Capote chain:

- **Convert** every (ABS X)+SCR X measurement to (ABS X)+SCA X — 26 conversions covering all transmission entries from baseline plus the step-5 additions.
- **Delete** the four Lemmel SCA–SCR datapoints (Nos. 89, 91, 93, 95).
- **Round-2 nuance** (see Appendix B): SCR(^{239}Pu) and SCR(^{241}Pu) become unconstrained and are removed from the fit parameter space, but SCR(^{233}U) and SCR(^{235}U) remain in the fit because the four reclassified entries from step 4 (ATOYI, WILLIS, FOOTE on U-235; GREEN-233 on U-233) now constitute *direct* SCR measurements. Roberto's original *"reducing 4 fitting variables SCR"* simplifies to *reducing 2 fitting variables*. Decision logged in B5.

Impact relative to step 5 Differences are small. SCA channel essentially unchanged (much less than 0.02%) — the SCA additions in step 4 are the dominant constraint. ABS, FIS, CA, and the F-quantities shift by ≤0.15% as the converted transmission datapoints now correlate only with SCA, slightly relaxing the absorption fit. χ^2 is comparable (80.5 → 80.1, a 0.4 *decrease*) with 2 fewer free parameters.

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	**13.3207**	2.883	**-0.524**
SCA 35	**14.0725**	1.305	**-0.345**
SCA 39	7.4716	5.105	-0.016

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 41	11.9950	6.665	-0.001
SCR 33	**12.6000**	5.556	**+1.781**
SCR 35	**12.9936**	3.172	**+1.723**
ABS 33	**572.7526**	0.185	**-0.128**
ABS 35	682.9085	0.180	-0.077
ABS 39	1020.1379	0.302	-0.054
ABS 41	1382.8937	1.047	-0.058
FIS 33	**527.9557**	0.239	**-0.127**
FIS 35	583.9164	0.207	-0.071
FIS 39	748.8023	0.301	-0.046
FIS 41	1018.0029	1.090	-0.054
CA 33	**44.7969**	1.806	**-0.146**
CA 35	**98.9921**	0.791	**-0.112**
CA 39	271.3356	0.855	-0.074
CA 41	364.8908	1.631	-0.069
CAP 34	95.9000	2.086	+0.000
NUB 33	2.4914	0.184	-0.001
NUB 35	2.4303	0.169	-0.002
NUB 39	2.8823	0.191	-0.003
NUB 41	2.9443	0.208	-0.002
NUB 52	3.7668	0.129	-0.001
ETA 33	2.2966	0.199	+0.000
ETA 35	2.0780	0.190	+0.004
ETA 39	2.1157	0.263	+0.005
ETA 41	2.1674	0.390	+0.002
ALPHA 33	0.0848	1.948	-0.019
ALPHA 35	0.1695	0.875	-0.042
ALPHA 39	0.3624	0.934	-0.028
ALPHA 41	0.3584	1.573	-0.015
WGA 33	1.0000	0.100	+0.000
WGA 35	0.9820	0.305	+0.000
WGA 39	1.0780	0.928	+0.000
WGA 41	1.0460	0.669	+0.000
WGF 33	0.9970	0.100	+0.000
WGF 35	0.9810	0.510	+0.000
WGF 39	1.0520	1.331	+0.000
WGF 41	1.0520	0.665	+0.000
FA 33	**572.7526**	0.210	**-0.128**
FA 35	670.6162	0.355	-0.077
FA 39	1099.7086	0.975	-0.054
FA 41	1446.5068	1.243	-0.058
FF 33	**526.3718**	0.259	**-0.127**
FF 35	572.8220	0.550	-0.071
FF 39	787.7400	1.364	-0.046
FF 41	1070.9390	1.277	-0.054
F1ETA 33	2.2897	0.244	+0.000
F1ETA 35	2.0759	0.624	+0.004
F1ETA 39	2.0647	1.643	+0.005
F1ETA 41	2.1799	1.021	+0.002
F2ETA 33	**1311.4128**	0.289	**-0.128**
F2ETA 35	1392.1472	0.569	-0.073
F2ETA 39	2270.5258	1.373	-0.049
F2ETA 41	3153.1933	1.275	-0.056
F3ETA 33	**738.6603**	0.443	**-0.128**
F3ETA 35	721.5310	1.098	-0.069
F3ETA 39	1170.8172	2.771	-0.044
F3ETA 41	1706.6865	1.824	-0.054

Step 5 vs Step 6: Head-to-Head Comparison

Both candidate fits agree to better than 0.2% on every primary and derived quantity. Step 6 has 2 fewer free parameters than step 5 (SCR(^{239}Pu), SCR(^{241}Pu) become unconstrained after the ABS+SCR → ABS+SCA conversion; SCR(^{233}U), SCR(^{235}U) remain in the fit, constrained by the round-2 reclassified entries — see Appendix B5). χ^2 is comparable: 80.55 vs 80.14, with step 6 *fitting slightly better* despite the parameter reduction.

Quantity	Step 5	Step 6	Δ%
SCA 33	13.3908	13.3207	**-0.524**
SCA 35	14.1213	14.0725	**-0.345**
SCA 39	7.4728	7.4716	-0.016

Quantity	Step 5	Step 6	Δ%
SCA 41	11.9951	11.9950	-0.001
SCR 33	12.3795	12.6000	**+1.781**
SCR 35	12.7734	12.9936	**+1.723**
SCR 39	6.6595	– (removed)	–
SCR 41	10.8374	– (removed)	–
ABS 33	573.4874	572.7526	**-0.128**
ABS 35	683.4317	682.9085	-0.077
ABS 39	1020.6859	1020.1379	-0.054
ABS 41	1383.6990	1382.8937	-0.058
FIS 33	528.6252	527.9557	**-0.127**
FIS 35	584.3285	583.9164	-0.071
FIS 39	749.1488	748.8023	-0.046
FIS 41	1018.5549	1018.0029	-0.054
CA 33	44.8622	44.7969	**-0.146**
CA 35	99.1032	98.9921	**-0.112**
CA 39	271.5372	271.3356	-0.074
CA 41	365.1441	364.8908	-0.069
CAP 34	95.9000	95.9000	+0.000
NUB 33	2.4914	2.4914	-0.001
NUB 35	2.4304	2.4303	-0.002
NUB 39	2.8824	2.8823	-0.003
NUB 41	2.9444	2.9443	-0.002
NUB 52	3.7668	3.7668	-0.001
ETA 33	2.2965	2.2966	+0.000
ETA 35	2.0780	2.0780	+0.004
ETA 39	2.1156	2.1157	+0.005
ETA 41	2.1674	2.1674	+0.002
ALPHA 33	0.0849	0.0848	-0.019
ALPHA 35	0.1696	0.1695	-0.042
ALPHA 39	0.3625	0.3624	-0.028
ALPHA 41	0.3585	0.3584	-0.015
WGA 33	1.0000	1.0000	+0.000
WGA 35	0.9820	0.9820	+0.000
WGA 39	1.0780	1.0780	+0.000
WGA 41	1.0460	1.0460	+0.000
WGF 33	0.9970	0.9970	+0.000
WGF 35	0.9810	0.9810	+0.000
WGF 39	1.0520	1.0520	+0.000
WGF 41	1.0520	1.0520	+0.000
FA 33	573.4874	572.7526	**-0.128**
FA 35	671.1299	670.6162	-0.077
FA 39	1100.2995	1099.7086	-0.054
FA 41	1447.3492	1446.5068	-0.058
FF 33	527.0393	526.3718	**-0.127**
FF 35	573.2262	572.8220	-0.071
FF 39	788.1045	787.7400	-0.046
FF 41	1071.5198	1070.9390	-0.054
F1ETA 33	2.2897	2.2897	+0.000
F1ETA 35	2.0758	2.0759	+0.004
F1ETA 39	2.0646	2.0647	+0.005
F1ETA 41	2.1798	2.1799	+0.002
F2ETA 33	1313.0907	1311.4128	**-0.128**
F2ETA 35	1393.1589	1392.1472	-0.073
F2ETA 39	2271.6351	2270.5258	-0.049
F2ETA 41	3154.9640	3153.1933	-0.056
F3ETA 33	739.6033	738.6603	**-0.128**
F3ETA 35	722.0290	721.5310	-0.069
F3ETA 39	1171.3357	1170.8172	-0.044
F3ETA 41	1707.6149	1706.6865	-0.054

Per the 2026-05-06 follow-up email, Step 5 is marginally favoured as the final fit candidate, on the grounds that the systematic SCR < SCA difference visible in step 5 is physically expected and yields fission cross sections in slightly better agreement with current library R-matrix fits. Differences with step 6 are within uncertainties.

K1 = F3ETA Evolution Through the Chain

K1 = F3ETA = NUB · FF − FA is the central derived quantity flagged as the K1 of the new TNC fit. Values across all 7 steps (round 2):

Step	F3ETA 33	F3ETA 35	F3ETA 39	F3ETA 41
step0 baseline	746.1581	723.5963	1169.1493	1710.5452
step1 lounsbury	741.9759	722.0181	1170.0323	1707.4469
step2 adamchuk	741.9650	721.8486	1169.9569	1707.3037
step3 wallner	741.9129	721.2836	1169.7702	1706.7079
step4 sca list	740.3842	722.0637	1170.3998	1706.9566
step5 transmission	739.6033	722.0290	1171.3357	1707.6149
step6 all sca	738.6603	721.5310	1170.8172	1706.6865

K1(^{233}U) drops from ≈ 746.2 (step 0) to ≈ 738.7 (step 6) — a 1.0% reduction driven primarily by the Lounsbury inclusion in step 1. K1(^{235}U) is essentially flat across the chain. K1(^{239}Pu) shifts ≈ −0.1% from step 0 to step 6. K1(^{241}Pu) is flat. The dominant K1 changes between round 1 and round 2 come from the Westcott-factor update (Appendix B4): U-235 K1 ≈ +0.3%, Pu-239 K1 ≈ −0.9%, Pu-241 K1 ≈ +1.1%.

Comprehensive Dataset Table (Final Fit, Step 6)

All 123 measurements actually used in the step-6 fit (after the microscopic filter is applied). Cross-references to the Decisions appendix flag entries where a non-trivial choice was made.

No.	Author	Reference	Quantity	Value	σ (%)	Appendix
5	LEMMEL	1982(1)	CAP 34	95.9	2.086	
21	LOUNSBURY	1970(1)	(CA 33)/FIS 33	0.0861	2.239	
22	LOUNSBURY	1970(1)	(CA 35)/FIS 35	0.1697	1.709	
23	LOUNSBURY	1970(1)	(CA 39)/FIS 39	0.3555	1.603	
54	DIVADEENAM	1984(2)	WGA 33	1	0.100	
55	DIVADEENAM	1984(2)	WGA 35	0.982	0.305	
56	DIVADEENAM	1984(2)	WGA 39	1.078	0.928	
57	DIVADEENAM	1984(2)	WGA 41	1.046	0.669	
58	DIVADEENAM	1984(2)	WGF 33	0.997	0.100	
59	DERUYTTER	1985(3)	WGF 35	0.981	0.510	
60	DIVADEENAM	1984(2)	WGF 39	1.052	1.331	
61	DIVADEENAM	1984(2)	WGF 41	1.052	0.665	
62	EGELSTAFF	1954(1)	(ABS 35)+SCA 35	724.33	3.590	
63	MELKONIAN	1953(1)	(ABS 35)+SCA 35	694.32	2.016	
64	PALEVSKY	1954(1)	(ABS 35)+SCA 35	700.32	1.428	
65	NITIKIN	1955(1)	(ABS 35)+SCA 35	710.32	2.956	
66	SIMPSON	1960(1)	(ABS 35)+SCA 35	690.32	1.391	
67	SAFFORD	1959(1)	(ABS 35)+SCA 35	696.32	0.359	
68	SAFFORD	1959(1)	(ABS 35)+SCA 35	698.32	0.730	
69	BLOCK	1960(1)	(ABS 35)+SCA 35	693.32	0.721	
70	SAPLAKOGLU	1961(1)	(ABS 35)+SCA 35	696.32	0.359	
71	GERASIMOV	1962(1)	ABS 35	670.31	2.089	
72	ZIMMERMAN	1955(1)	(ABS 39)+SCA 39	1022.5	1.271	
73	NITIKIN	1955(1)	(ABS 39)+SCA 39	1040.5	2.979	
74	BOLLINGER	1958(1)	(ABS 39)+SCA 39	1022.5	1.369	
75	PATTENDEN	A1956(1)	(ABS 39)+SCA 39	1012.5	3.003	
76	SAFFORD	1961(1)	(ABS 39)+SCA 39	1018.5	0.727	
77	MUETHER	1954(1)	(ABS 33)+SCA 33	597.27	2.344	
78	NITIKIN	1955(1)	(ABS 33)+SCA 33	577.26	3.638	A8
79	PATTENDEN	1956(1)	(ABS 33)+SCA 33	596.27	3.019	
80	GREEN	1957(1)	ABS 33	574.26	3.483	
81	SIMPSON	1960(1)	(ABS 33)+SCA 33	587.27	0.800	
82	SAFFORD	1960(1)	(ABS 33)+SCA 33	585.77	0.990	
83	SAFFORD	1960(1)	(ABS 33)+SCA 33	585.67	0.410	A9
84	BLOCK	1960(1)	(ABS 33)+SCA 33	587.27	0.511	
85	SIMPSON	1961(1)	(ABS 41)+SCA 41	1386.6	3.606	
86	CRAIG	1964(1)	(ABS 41)+SCA 41	1380.6	2.897	
87	SMITH	1984(3)	(ABS 41)+SCA 41	1395.6	1.433	
96	DERUYTTER	1974(1)	39 FH1 39	1812.1	0.454	
97	DERUYTTER	1973(1)	34 FH1 35	1439.1	0.445	
98	DERUYTTER	1961(1)	FIS 35	589.37	1.325	
99	WHITE	1967(1)	(FIS 39)/FIS 35	1.253	1.800	
100	WHITE	1967(1)	(FIS 41)/FIS 35	1.763	2.751	
101	BERCEANO	1977(1)	34 FH1 35	1443	0.791	
102	RAFFLE	1959(1)	33 FH1 33	834.26	3.302	
103	RAFFLE	1959(1)	FIS 35	581.97	3.092	
104	RAFFLE	1959(1)	39 FH1 39	1714.8	2.843	
105	RAFFLE	1956(1)	(FIS 41)/FIS 39	1.3319	6.000	
106	MASLIN	1965(1)	FIS 35	583.71	1.449	

No.	Author	Reference	Quantity	Value	σ (%)	Appendix
107	SAPLAKOGLU	1959(1)	FIS 35	593.17	2.232	
108	WATANABE	1964(1)	FIS 41	985.45	4.569	
109	FRAYSSE	1965(1)	(39 FH1 39)/34 FH1 35	1.2425	1.557	
110	BORCEA	1970(1)	(39 FH1 39)/34 FH1 35	1.2578	1.624	
111	BOLDEMAN	1977(1)	NUB 52	3.7549	0.431	
112	BOLDEMAN	1980(1)	(NUB 33)/NUB 52	0.66049	0.255	
113	BOLDEMAN	1980(1)	(NUB 35)/NUB 52	0.64318	0.255	
114	BOLDEMAN	1980(1)	(NUB 39)/NUB 52	0.76614	0.243	
115	BOLDEMAN	1980(1)	(NUB 41)/NUB 52	0.78028	0.231	
116	SPENCER	1982(1)	NUB 52	3.7831	0.221	
117	GWIN	1984(1)	(NUB 33)/NUB 52	0.65996	0.340	
118	GWIN	1984(1)	(NUB 35)/NUB 52	0.64701	0.265	
119	GWIN	1984(1)	(NUB 39)/NUB 52	0.7653	0.257	
120	GWIN	1984(1)	(NUB 41)/NUB 52	0.78423	0.290	
121	HOPKINS	1963(1)	NUB 52	3.7767	0.838	
122	HOPKINS	1963(1)	(NUB 33)/NUB 52	0.65483	0.895	
123	HOPKINS	1963(1)	(NUB 35)/NUB 52	0.64587	0.838	
124	HOPKINS	1963(1)	(NUB 39)/NUB 52	0.75047	0.996	
125	ASPLUND	1963(1)	NUB 52	3.791	1.066	
126	CONDE	1965(1)	(NUB 35)/NUB 52	0.64222	0.847	
127	WHITE	1968(1)	NUB 52	3.8194	1.033	
128	AXTON	A1985(3)	NUB 52	3.7547	0.300	
129	COLV/AXT	1966(1)	NUB 52	3.7299	0.806	
130	COLV/ULL	1965(1)	NUB 52	3.7405	0.438	
131	COLVIN	1965(1)	(NUB 33)/NUB 35	1.0204	0.587	
132	COLVIN	1965(1)	(NUB 39)/NUB 35	1.1835	0.675	
133	COLVIN	1965(1)	(NUB 41)/NUB 35	1.2102	0.905	
134	COLVIN	1965(1)	(NUB 35)/NUB 52	0.64334	0.472	
135	MATHER	1964(1)	(NUB 33)/NUB 52	0.67093	1.235	
136	MATHER	1964(1)	(NUB 35)/NUB 52	0.64145	0.537	
137	MATHER	1964(1)	(NUB 39)/NUB 52	0.77307	1.122	
138	ALEK-ROV	1981(1)	NUB 52	3.7618	0.483	
139	SMITH	1984(1)	NUB 52	3.7678	0.303	
140	EDWARDS	1982(1)	NUB 52	3.7641	0.711	
141	BOZ-NESH	1977(1)	NUB 52	3.7475	0.580	
142	DEVOLPI	1972(1)	NUB 52	3.7507	0.463	
143	ZHANG	1981(1)	NUB 52	3.7534	0.490	
144	SPIEGEL	1981(1)	NUB 52	3.7828	0.759	
145	SMITH	1984(1)	ETA 33	2.2972	0.350	
146	SMITH	1984(1)	ETA 35	2.0842	0.386	
147	SMITH	1984(1)	ETA 39	2.112	0.380	
148	SMITH	1984(1)	ETA 41	2.169	0.419	
149	MACKLIN	1960(1)	ETA 33	2.3121	0.375	
150	MACKLIN	1960(1)	ETA 35	2.0757	0.382	
151	MACKLIN	1962(1)	ETA 39	2.1163	0.851	
165	LLOYD	1982(1)	(ETA 39)/FIS 39	0.002782	2.072	
166	SPENCER	1985(3)	(ABS 35)+SCA 35	690	1.000	A1
167	SPENCER	1985(3)	(ABS 39)+SCA 39	1025.8	1.000	A1
168	ADAMCHUK	At. En. (R), 65, 434 (1988)	(CA 35)/FIS 35	0.169	2.100	
169	WALLNER	PRL 112, 192501 (2014)	CA 35	99.5	2.000	
170	WALLNER	PRL 112, 192501 (2014)	CA 35	101.5	2.000	
171	ATOYI	Acta Cryst. 20, 587 (1966)	SCR 35	12.3	4.065	B2
172	WILLIS	PRS A274, 122 (1963)	SCR 35	12.8	7.812	B2
174	ARIF	PR A35, 2810 (1987)	SCA 35	14	1.429	
175	CEULEMANS	Divadeenam compilation 1970	SCA 35	14.6	3.425	A4
176	BAYVAS	Divadeenam compilation 1980	SCA 35	14.3	10.489	
177	FOOTE	Divadeenam compilation 1958	SCR 35	15	6.667	B3
178	OLEKSA	Phys Rev 109, 1645 (1958)	SCA 33	13.4	3.731	A5
179	GREEN	JNM 51, 281 (1974)	SCR 33	12.6	5.556	B2
180	VERTEBNYI	Divadeenam compilation 1974	SCA 33	13.2	4.545	
181	GREEN	JNM 34, 281 (1970)	SCA 39	7.7	7.792	
182	ATOYI	Acta Cryst. 20, 587 (1966)	SCA 39	7.3	9.589	
183	ROOF	Acta Cryst. 15, 351 (1962)	SCA 39	7.3	9.589	
184	PRONYAEV	Internal analysis 2026-04-17	SCA 41	12	6.667	A7
185	ANTONOV	Capote 2026-05-04 list	(ABS 35)+SCA 35	695	2.158	A10
186	GERASIMOV	Capote 2026-05-04 list	(ABS 35)+SCA 35	685	1.314	A11
187	LEONARD	Capote 2026-05-04 list (1954)	(ABS 35)+SCA 35	702	1.994	A12
188	PSCHENICHNYI	Capote 2026-05-04 list (1978)	(ABS 33)+SCA 33	587.9	0.510	A13
189	HARVEY-1	Capote 2026-05-04 list (1979)	(ABS 33)+SCA 33	585	0.239	A14
190	HARVEY-2	Capote 2026-05-04 list (1979)	(ABS 33)+SCA 33	592	1.014	A14
191	SCHWARTZ	Capote 2026-05-04 list (1958)	(ABS 41)+SCA 41	1410	5.674	
192	ANDERSON	Capote 2026-05-04 list (1945)	(ABS 39)+SCA 39	1045	2.392	
193	PALEVSKY	Capote 2026-05-04 list (1955)	(ABS 39)+SCA 39	1025	0.976	

No.	Author	Reference	Quantity	Value	σ (%)	Appendix
194	LEONARD	Capote 2026-05-04 list (1956)	(ABS 39)+SCA 39	1055	1.232	A15

A. Decisions and Contentious Points

Every choice that involved interpretation, departure from a literal reading of Roberto's emails, or risk of double-counting is recorded here so collaborators can scrutinise it.

A1. Spencer 1987 vs. existing No.166/167 — values used as-is

Roberto's "Revision/additions" email lists Spencer 1987 ABS+SCR(^{235}U) = 690(5) and Spencer 1987 ABS+SCR(^{239}Pu) = 1025(6). The baseline already contains:

- No.166 SPENCER 1985(3) ABS+SCR(^{235}U) = 690.0, σ = 1.0% (= 6.9 b), with a single systematic component `alpha = 0.707`.
- No.167 SPENCER 1985(3) ABS+SCR(^{239}Pu) = 1025.8, σ = 1.0% (= 10.26 b), same systematic.

The values match within rounding (690 vs 690.0; 1025 vs 1025.8 — likely the same measurement, presented at a 1985 conference and published 1987). Per project decision, the database entries are retained as-is rather than being re-overwritten with Roberto's values. **Note**: the uncertainties differ noticeably (DB: 1.0%; Roberto: 0.72% / 0.59%); the more conservative DB uncertainty is what enters the fit. Step-6 conversion turns these into ABS+SCA(^{235}U) and ABS+SCA(^{239}Pu) like all other ABS+SCR entries.

A2. Atoyi 1966 ABS+SCR tag treated as SCA

In the SEARS column of Capote's 2026-04-30 inlined compilation, the Atoyi 1966 row is annotated "SCR". The same row also gives Sears Sel = 12.3(5) b. Per project decision, the entire compilation is treated as providing SCA values regardless of the per-row "SCR/SCA" tag, because (a) Sears' formalism delivers the bound-atom S_el (= SCA) by construction, and (b) the per-row tags in Roberto's table show inconsistencies that are most plausibly explained as paste artifacts from a heterogeneous source table.

A3. Willis 1963 — EXFOR value used

The Sears compilation gives Willis 1963 = 12.1(5) coh + 0.2(2) inc ⟶ S_el 12.3(5). Roberto annotates "replace with EXFOR value" and the Divadeenam section gives "WILLIS 1963 12.8(1.0) = EXFOR". The EXFOR value 12.8(1.0) is used in step 4.

A4. Ceulemans 1970 — V-51 renormalisation used as-is

Roberto's table notes "EXFOR 14.3(5) rel V-51 (1.02)" — the 14.6(5) value used in step 4 is 14.3 × 1.02, the 1.02 factor being a renormalisation against vanadium-51 standard. Different evaluations of V-51 thermal scattering vary by ~1–3% (Hughes-Schwartz 1958: ~5.0 b; Sears 1992 / ENDF/B-VIII: 5.08 b; ratio ≈ 1.016). Re-renormalising against current standards would change Ceulemans by < 1% (≈ 0.05–0.15 b), well under the quoted σ = 0.5 b — no practical impact on the fit. **Used as-is per project decision.**

A5. Oleksa 1960 — quoted at 0.1 eV, used as-is

The EXFOR replacement Oleksa 13.4(5) is quoted "at 0.1 eV", not at the thermal point 0.0253 eV. Two candidate corrections were estimated:

- **Bound⟶free transition**: σ_free / σ_bound = $(A/(A+1))^2$ = 0.9914 for A=233, i.e. <0.3% effect at 0.1 eV (still in bound regime); ≤ 0.04 b.
- **Resonance contribution**: U-233 has no large resonance below 1 eV. Tail contributions from negative resonances and the 1.45 eV resonance at 0.1 eV are at most ≈ 0.05 b.

Combined $|\Delta\sigma| < 0.1$ b, well below quoted σ = 0.5 b. Used as-is per project decision.

A6. Green 1974 ABS+SCR tag treated as SCA

Same situation as A2: the Divadeenam-section row for Green 1974 carries a "SCR" annotation but is added as SCA(^{233}U) per the global decision to treat all compilation rows as SCA.

A7. Pu-241 SCA — Pronyaev internal analysis

No direct experimental SCA measurement exists for ^{241}Pu. Pronyaev's 2026-04-17 internal analysis derives SCA(^{241}Pu) = 12.0(0.8) b from the potential scattering radius R_pot = 9.54 fm (consistent across ENDF/B-VIII.1, JEFF-4.0 and JENDL-5; $4\pi R^2$ = 11.44 b plus the contribution of negative resonances in the dominant ENDF/B-VIII.1 evaluation). This estimate is encoded as a direct SCA(^{241}Pu)

"measurement" replacing the deleted Lemmel No.94 (12.006 b, σ = 21.6%). The replacement reduces the SCA(^{241}Pu) uncertainty from 21.6% (Lemmel) to 6.7% (Pronyaev).

A8. Nikitin 1955 spelling — DB uses "NITIKIN"

Roberto writes "Nikitin" (correct transliteration of the Russian Никитин); the database has the typo "NITIKIN" inherited from Axton 1986 throughout (Nos. 65, 73, 78). The DB spelling is preserved for stable cross-referencing.

A9. Safford 1960 vs 1959 ambiguity

Roberto's "Revision/additions" email instructs *"Change Safford 1959 ABS+SCR to ABS+SCA"* for U-233. The database has no Safford 1959 U-233 entry; Safford's U-233 transmission measurements are dated 1960 (Nos. 82 ABS+SCA(^{233}U) = 585.77 already, 83 ABS+SCR(^{233}U) = 585.67). The 1959 → 1960 mismatch is treated as a typo and No.83 is reclassified per Roberto's evident intent. Worth verifying with Roberto if there is an actual 1959 paper we are missing.

A10. Antonov 1986 ABS+SCA tag — non-default for transmission

Roberto tags Antonov 1986 ^{235}U as ABS+SCA, which is unusual for a transmission measurement (the dominant convention is ABS+SCR for rolled-metal samples). The tag is honoured under the assumption that Antonov used a non-metallic sample (UO_2 powder?) or applied a proper extinction correction. Verify with Roberto / read the original paper.

A11. Gerasimov 1962 — likely separate from existing No.71

A GERASIMOV 1962 entry exists in the baseline as **No.71 GERASIMOV 1962(1) ABS 35 = 670.31 b, σ = 2.09%** (bare ABS, no scatter component). Roberto's "Gerasimov 62 ABS+SCR 685(9)" has a different value (685 vs 670.3) and a different quantity (ABS+SCR vs ABS). Per project decision, this is added as a separate measurement (No.186) rather than overwriting No.71. Worth verifying with Roberto if this is in fact the same measurement re-analysed.

A12. Leonard 1954 — type defaulted to ABS+SCR

Roberto's email gives "Leonard 54 702(14)" with no SCR/SCA tag. Defaulted to ABS+SCR per the convention for the era (rolled-metal samples) and pending paper lookup. This decision is **moot for the final fit** because step 6 converts all ABS+SCR to ABS+SCA in any case.

A13. Pschenichnyi 1978 uncertainty — typo fix 0.3 → 3.0

Roberto's email reads *"Pschenichnyi 78 ABS+SCR 587.9(3)"*, which would be σ = 0.3 b on 587.9 b ≈ 0.05% — implausibly small for a transmission measurement. **Treated as a typo and used as σ = 3.0 b** (≈ 0.51%). This decision is conservative; if Roberto truly intended 0.3 b, Pschenichnyi would dominate the U-233 absorption fit by orders of magnitude.

A14. Harvey-1 / Harvey-2 — kept uncorrelated; correlation alternative noted

The two Harvey 1979 ABS+SCR(^{233}U) entries come from the same author, same year, and likely the same ORELA campaign — physical correlation through sample mass / normalisation is plausible. Per project decision, the two are entered with empty `systematic_uncertainties` and treated as independent, matching the convention for nearly all other Axton transmission entries.

Alternative considered (not adopted): introduce a shared code `harvey79` with a 0.20% common component:

```
No.189 Harvey-1: σ_total = 0.24%; sys = {harvey79: 0.20}
No.190 Harvey-2: σ_total = 1.01%; sys = {harvey79: 0.20}
```

This would marginally tighten the U-233 ABS uncertainty. Worth revisiting if Roberto wants tighter modelling of correlated systematics.

A15. Leonard 1956 ABS+SCR(^{239}Pu) — largest tension in step 5

Leonard 1956 ABS+SCR(^{239}Pu) = 1055 b lies +2.1σ above the fit-predicted 1027.4 b. This contributes ≈4.5 of the 9.5 χ^2 increase between step 4 and step 5. The measurement is retained as-is; flagged here as the dominant transmission tension in the chain.

A16. SCR removed from parameter space in step 6

Roberto's step 6 instructs *"CONVERT ALL SCR in ABS+SCR to ABS+SCA, delete Lemmel SCA-SCR (reducing 4 fitting variables SCR)"*. The literal application:

- Convert every (ABS X)+SCR X ⟶ (ABS X)+SCA X (26 conversions).
- Delete the four Lemmel SCA–SCR datapoints (Nos. 89, 91, 93, 95).
- **No measurement function or datapoint references SCR after these changes**, so the four SCR primary parameters become unconstrained.
- They are therefore removed from the fit parameter space, consistent with the existing handling of unconstrained parameters (CA 40, CA 42, GC116, GA116). This matches Roberto's parenthetical "reducing 4 fitting variables SCR" exactly.

A17. Microscopic filter and unconstrained parameters

All seven benchmarks apply the same microscopic filter that reproduces Axton Table 8: exclude measurements whose `measured_function` references any Maxwellian-derived quantity (FA, FF, CAP, FFH, F1ETA, F2ETA, F3ETA, FLEM, F1BIG, F1CAB–F5CAB, GC116, GA116) — except CAP 34 (No.5), which is retained because it ≈ σ_γ for non-fissile U-234. Lounsbury alpha (Nos. 21–23) is excluded only in step 0; it is included in steps 1–6. Unconstrained parameters CA 40, CA 42, GC116/40/41/42, GA116/39/41 are removed from the fit parameter space throughout. Step 6 additionally removes SCR 33/35/39/41 (see A16).

A18. Empty systematic uncertainties for step-5 transmission additions

All step-5 transmission additions are entered with empty `systematic_uncertainties` (single uncorrelated total uncertainty), matching the convention used for 26 of the 28 baseline transmission entries. Spencer 1985 (Nos. 166/167) is the existing exception with `alpha = 0.707` representing the α-correction Spencer used internally. The Harvey-pair correlation alternative is the only other systematic structure considered (see A14).

B. Round-2 Corrections (2026-05-08)

The 2026-05-08 email round produced three corrections to the round-1 fit. Each is documented below; the verbatim source is preserved in `references/emails/tnc_update_roberto_20260508/summary.md`. The original round-1 outputs are preserved as `evaluation/capote_chain_summary_20260504.json` and `evaluation/report_capote_fit_20260504.pdf` for direct numerical comparison.

B1. KAISER (Physica B 136, 134 (1986)) removed as duplicate of ARIF

Kaiser et al. 1986 and Arif et al. 1987 used the same neutron-interferometry installation at the same institute; Kaiser gave preliminary values at only two energies, while Arif is the final publication. The 2026-05-08 review (Pronyaev) flagged the duplication; the 2026-05-05 follow-up confirmed deletion of KAISER, retaining ARIF.

B2. ATOYI / WILLIS / GREEN-233 reclassified SCA ⟶ SCR

In round 1, the entire Sears/Divadeenam compilation was treated as providing SCA values per project decision (the per-row "SCR" tags were assumed to be paste artefacts). The 2026-05-08 review identified three rows for which the SCR tag is consistent with the actual measurement method:

- **ATOYI 1966** (^{235}U, Sears Sel = 12.3(5)): tagged SCR by Sears.
- **WILLIS 1963** (^{235}U, EXFOR value 12.8(1.0)): SCR per Sears.
- **GREEN 1974** (^{233}U, 12.6(7)): per the Pronyaev analysis, the EXFOR record indicates a classical (non-crystal-spectrometer) method, consistent with SCR. Ad-hoc SCA estimation supplied by Pronyaev.

All three are reclassified from SCA to SCR in steps 4, 5 and 6.

B3. FOOTE 1958 reclassified SCA ⟶ SCR (inferred)

The cleaned compilation in Roberto's 2026-05-05 email tags FOOTE 1958 (^{235}U, 15.0(1.0)) as "SCR", but the explicit numbered action items only list ATOYI, WILLIS and GREEN-233. Per project decision, the cleaned table is treated as the new authoritative classification, so FOOTE is reclassified to SCR alongside the three named entries. **This decision is inferred and worth confirming with Roberto**. The K1 impact of this single reclassification is ≤ 0.05% (FOOTE's 6.7% uncertainty is much weaker than ARIF's 1.4% which dominates the SCA(^{235}U) channel).

B4. Westcott factors updated to B81-derived values

The 2026-05-06 email replaces the Westcott factors with values derived from the B81 evaluation (R-matrix fit best estimate), with uncertainties set by the spread between B81 and JEFF-4.0:

nuclide	Old WGA (Gabs)	New WGA (Gabs)	Old WGF (Gf)	New WGF (Gf)
U-233	0.9988 (0.12%)	**1.000 (0.10%)**	0.9966 (0.20%)	**0.997 (0.10%)**
U-235	0.9787 (0.09%)	**0.982 (0.31%)**	0.9778 (0.09%)	**0.981 (0.51%)**
Pu-239	1.0767 (0.27%)	**1.078 (0.93%)**	1.0562 (0.28%)	**1.052 (1.33%)**
Pu-241	1.0444 (0.19%)	**1.046 (0.67%)**	1.0452 (0.67%)	**1.052 (0.67%)**

Gabs is derived as Gabs = (Gcap·Scap + Gfis·Sfis)/(Scap+Sfis) per Capote's 2026-05-06 derivation. The new factors apply to **all seven steps**: steps 0 and 1 read from a new `axton1986_data_capote_round2_baseline.json` file (= original baseline + new Westcott); steps 2–6 inherit through their respective JSONs.

K1 impact (Step 6, round 2 vs round 1):

Nuclide	K1 (round 1)	K1 (round 2)	Δ
^{233}U	738.81	738.66	-0.020%
^{235}U	719.32	721.53	+0.307%
^{239}Pu	1181.26	1170.82	-0.884%
^{241}Pu	1688.54	1706.69	+1.075%

For comparison, K1(^{235}U, Step 5, round 2) = 722.03, in close agreement with the value reported in the 2026-05-06 email (722.4) using the same new Westcott factor.

B5. SCR(^{233}U), SCR(^{235}U) retained as fit variables in step 6

In round 1, step 6 removed all four SCR primary parameters from the fit because no measurement function or datapoint referenced SCR after the conversions and Lemmel deletion. In round 2, the reclassifications (B2, B3) introduce four direct SCR measurements: ATOYI (^{235}U), WILLIS (^{235}U), FOOTE (^{235}U), and GREEN-233 (^{233}U). These constrain SCR(^{233}U) and SCR(^{235}U), so the two parameters remain in the fit. SCR(^{239}Pu) and SCR(^{241}Pu) have no remaining constraints and stay unconstrained (removed).

Net effect: Roberto's parenthetical *"reducing 4 fitting variables SCR"* in the original step 6 description simplifies to *reducing 2 fitting variables*. Step 6 now has 28 free parameters (was 26 in round 1).

B6. Round-2 baseline file separation

To preserve the Axton-Table-8 reproduction artefact, the original `axton1986/axton1986_data.json` is left **unchanged** and a new file `axton1986/axton1986_data_capote_round2_baseline.json` is introduced for steps 0 and 1. The new file is identical to the original except for the eight Westcott measurements (No. 54–61). Steps 2–6 carry their own copies of all measurements; the Westcott updates are applied in-place to those JSONs.

C. Files Produced by This Chain

All on branch `tnc_update_capote_20260504`, building on commit `575fac0` (the Axton 1986 reproduction state of master).

JSON databases (one per step where data changes)

- `axton1986/axton1986_data.json` — original Axton-Table-8 reproduction baseline (untouched; not currently used by any step)
- `axton1986/axton1986_data_capote_round2_baseline.json` — round-2 baseline (= original + new Westcott); steps 0 and 1 read this
- `axton1986/axton1986_data_capote_step2_adamchuk.json` — 168 measurements
- `axton1986/axton1986_data_capote_step3_wallner.json` — 170 measurements
- `axton1986/axton1986_data_capote_step4_sca_list.json` — 179 measurements (round 2: 180 − 1 KAISER deleted)
- `axton1986/axton1986_data_capote_step5_transmission.json` — 189 measurements (round 2: 190 − 1 KAISER deleted)
- `axton1986/axton1986_data_capote_step6_all_sca.json` — 185 measurements (round 2; final)

Benchmarks

- `evaluation/benchmark_capote_step0_baseline.py`
- `evaluation/benchmark_capote_step1_lounsbury.py`
- `evaluation/benchmark_capote_step2_adamchuk.py`
- `evaluation/benchmark_capote_step3_wallner.py`
- `evaluation/benchmark_capote_step4_sca_list.py`
- `evaluation/benchmark_capote_step5_transmission.py`
- `evaluation/benchmark_capote_step6_all_sca.py`
- `evaluation/benchmark_capote_summary.py` — runs all 7 in sequence and dumps `capote_chain_summary.json`

Generated artefacts

- `evaluation/capote_chain_summary.json` — machine-readable consolidation of all step results (round 2)
- `evaluation/capote_chain_summary_20260504.json` — round-1 snapshot for direct comparison
- `evaluation/report_capote_fit.md` — this report (round 2)
- `evaluation/report_capote_fit_20260504.pdf` — round-1 PDF snapshot
- `evaluation/build_capote_report.py` — the script that generated this report from the summary JSON

Source paraphrases

- `references/emails/tnc_update_roberto_20260505/summary.md` — paraphrased thread covering the round-1 instructions (2026-04-09 → 2026-05-04)
- `references/emails/tnc_update_roberto_20260508/summary.md` — paraphrased round-2 corrections (2026-05-05 → 2026-05-06)